

Salient Characteristics

USBC-Certified String Pinsetter System

1. General Requirement

The Government requires a string pinsetter system that is fully compatible with the existing bowling center management system, Conqueror X, to ensure centralized control, operational efficiency, and compliance with governing standards.

2. Salient Characteristics

To be considered technically acceptable, the proposed system **shall meet or exceed** the following characteristics:

2.1 Native System Integration

- Shall provide **direct, native integration** with the Conqueror POS system
 - Shall not rely on third-party middleware, custom-built interfaces, or standalone control systems for core functionality
 - Shall support real-time communication between:
 - Point-of-sale (POS)
 - Scoring system
 - Pinsetter system
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2.2 Centralized Lane Control

- Shall allow the management system to:
 - Control lane assignment and status (e.g., open play, league, tournament)
 - Initiate and terminate bowling sessions
- Shall eliminate the need for manual control inputs at the lane or pinsetter level

2.3 Automated Gameplay Configuration

- Shall support **system-controlled application of gameplay modes** based on session type
- Shall include, at a minimum:
 - Competitive (USBC-compliant) mode in accordance with United States Bowling Congress standards
 - Recreational/entertainment mode
- Transition between modes shall:
 - Be initiated through the management system
 - Not require manual mechanical adjustment
 - Not require lane shutdown or service interruption

2.4 Bi-Directional Data Communication

- Shall provide **two-way communication** between the pinsetter system and management system
- Shall transmit, at a minimum:
 - Lane status (active, idle, fault)
 - Active gameplay mode
 - Session/game data
- Shall allow the management system to:
 - Receive system feedback in real time
 - Log and report operational data

2.5 Compatibility with Existing Infrastructure

- Shall be compatible with:
 - Existing scoring systems
 - Current lane configurations and infrastructure

- Shall not require significant structural modification to the facility
 - Shall minimize additional hardware requirements for integration
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2.6 Automation & Labor Reduction

- Shall reduce or eliminate the need for:
 - Manual lane configuration
 - Physical adjustment of pinsetter settings
 - Separate operator interfaces for pinsetter control
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2.7 Reliability of Integration

- Shall maintain stable operation without:
 - Frequent communication loss
 - System desynchronization between POS and pinsetter
 - Shall not introduce additional single points of failure through external interfaces
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3. Acceptability Criteria

A system will be considered acceptable if it demonstrates:

- Full functional integration with the existing management system
- Automated control of gameplay modes without manual intervention
- Reliable, real-time communication between all subsystems
- Capability to support both competitive and recreational operations seamlessly